Problem FoL.1 ... tough run

With what velocity should you run at the equator in order to weigh as much as possible (e.g
have the maximum possible weight, not mass) if you can choose the optimal direction?

Matěj's healthy method of gaining weigh

Problem FoL.2 ... electrobell

Mirek's colleague bought a \$1 *"small science" apparatus at a flea market. It's a small box wit two wires. The ends of these wires are almost connected, leaving a small distance l betwee them. If we connect the wires to a powerful voltage source, we can create a short electric arc The sound of this discharge is quite loud, that's why Mirek's colleague uses this device a lunch bell. Find the largest possible distance (**in micrometres**) of the wires that allows th creation of an electric discharge for a source with peak voltage* $325\,\mathrm{V}$. *The dielectric strengt of air is* $D = 3\,\mathrm{MV{\cdot}m^{-1}}$.

Mirek was looking forward to lunch

Problem FoL.3 ... jump

The mass of the average human is 80 kg. *How many people would need to gather in one plac*
and jump 1 m *up at the same time in order to shift Earth's center by* 0.1 pm*?*

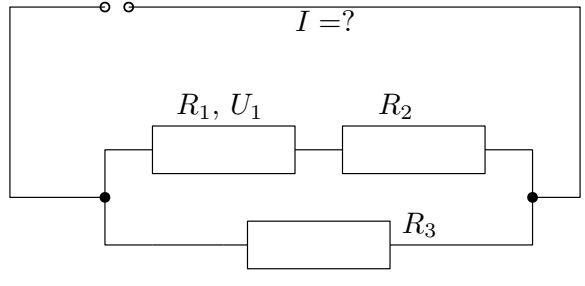
Matěj was jumping on a trampoline

Problem FoL.4 ... there's a current flowing

What is the total current I flowing through the depicted circuit? Resistances in the diagram are $R_1 = 124\,\Omega$, $R_2 = 263\,\Omega$ and $R_3 = 454\,\Omega$. Voltage on the first resistor is $U_1 = 14.8\,\mathrm{V}$.

Enter the result in miliamperes.

Karel wanted you to review electrical circuits.



Problem FoL.5 ... a mirror problem

There's a 10 m *wide,* 5 m *long rectangular room. In the middle of one of the longer walls, there'*
a mirror with width 1 m *at eye level. Mikuláš is standing in front of the middle of the mirror a*
a distance 1 m *from the mirror. What part of the room's area (in %) will he see in the mirror*
Neglect the fact that part of this area won't be visible because he sees himself in front of it
Note: when computing the area, consider only the horizontal cross-section of the room at ey
level. *Katka was looking out of a window during a lecture*

Problem FoL.6 ... supercooled water

There's a container with supercooled water at temperature $t = -8\,^{\circ}\mathrm{C}$. *Initially there are n condensation nuclei, so it remains in the liquid state. Find the mass percentage of this liquid tha will freeze after inserting a condensation nucleus. Neglect the heat capacity of the containe Use the specific heat capacity of water* $c = 4{,}180\,\mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}}$ *and the latent heat of fusio* $= 334\,\mathrm{kJ{\cdot}kg^{-1}}$.

Remember: This is an easy problem, so keep it simple.

Karel attended a talk by doc. Bochníček on the properties of supercooled wate

Problem FoL.7 ... ecological

Imagine that electricity could only be produced by burning wood. Let's assume we have jus enough wood to produce the amount of electricity necessary to power a tablet for one hou How many sheets of paper could be made from this wood instead of burning it? The heatin value of wood is $13\,\mathrm{MJ\cdot kg^{-1}}$. *During production, distribution and storage of electricity,* 80 % *of the energy (heat) is lost. The tablet is fueled by a* $3.6\,\mathrm{V}$ *battery with current consumptio of* $0.5\,\mathrm{A}$. *One sheet of paper weighs* $5.0\,\mathrm{g}$ *and double of this mass in wood is needed for it production.*

Erik is saving the forests

Problem FoL.8 ... headshot

We fire a projectile with mass $m = 0.502\,\mathrm{g}$ *from an air gun with mass* $M_\mathrm{p} = 5\,\mathrm{kg}$*. The projectil mpacts a cuboid with mass* $M = 182\,\mathrm{g}$*, which slides and stops at distance* $s = 4.8\,\mathrm{cm}$*. Th riction coefficient between the cuboid and surface is* $f = 0.3$*. Determine the muzzle velocity o he projectile.*

Assume that the velocity of the projectile at the moment of impact is the same as the muzzl elocity and that the cuboid is far enough to be unaffected by gases emitted from the air gun The cuboid lies on a horizontal surface.

Adapted by Karel from the article Fyzika jako zážitek ("Physics as an experience")

Problem FoL.9 ... transportation acceleration

So one day Karel was casually riding the Prague metro and saw an information panel abou rams. It said that the power of a tram car increased tenfold in the last decades. How man imes did the maximum speed increase, assuming their weight didn't change and the resistiv orce are proportional to the second power of tram's velocity? In other words, we want to kno he value of $k = v_1/v_0$, where v_0 is the original maximum speed and v_1 is the current one.

Karel saw a tram praising ad in the Prague metro

Problem FoL.10 ... closer than it seems

According to ground-based observatories on Earth, the parallax of the star Proxima Centau
is $p = 0.77\,\mathrm{arcsec}$. There's a planet orbiting this star at distance $r = 0.05\,\mathrm{au}$ (assume its orbi
is approximately circular). Let's imagine there are intelligent beings living on this planet (let'
call them Centaurs) and they measured the parallax of our Sun. The Centaurian definition o
the parallax is, of course, based on their home planet's orbit. How large is the parallax the
measured? Enter the result in **arcseconds**.

Mirek was pondering about extraterrestrial physical units

Problem FoL.11 ... soda

How many times would the volume of a soda increase if all dissolved carbon dioxide suddenl
urned to gas? A typical soda contains $8\,\mathrm{g\cdot dm^{-3}}$ *of carbon dioxide. Consider the situation a*
$5\,^{\circ}\mathrm{C}$, $101.3\,\mathrm{kPa}$. *Štěpán spilled his drink on himself*

Problem FoL.12 ... don't fall!

The two-wheeled vehicle segway maintains its vertical orientation by accelerating or deceleratng. Neglect the mass of the vehicle and consider the driver a point with mass $65\,\mathrm{kg}$ *at a distanc* $\ell = 1\,\mathrm{m}$ *from the axis of rotation. What power does the motor need to provide when the segwaneeds to balance a tilt by* $\alpha = 10^\circ$ *to the front at speed* $v = 10\,\mathrm{km{\cdot}h^{-1}}$*? Use* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.*

Michal was wondering about the new road signs

Problem FoL.13 ... shoot it down!

What's the probability (in percent) that if we shine a laser vertically upwards for a short time
he ray hits a passenger airplane? The average horizontal cross-section of an airplane is S =
= 300 m^2 and at each moment, there are approximately ten thousand airplanes in the air
Assume that airplanes are distributed homogeneously across the sky and fly at heights much
maller than the radius of Earth. *Matěj read about a terror attack*

Problem FoL.14 ... Young's thermal stress

A copper rod of length $l = 12.3\,\mathrm{cm}$ *and cross-sectional area* $S = 1.02\,\mathrm{cm}^2$ *is fixed so that it length cannot thermally increase. The linear coefficient of thermal expansion for copper is* $\alpha = 1.70 \cdot 10^{-5}\,\mathrm{K}^{-1}$ *and Young's elastic modulus is* $E = 117\,\mathrm{GPa}$. *Find the force exerted by the rod on the holder due to thermal expansion. Give the result for* $\Delta T = 15.0\,\mathrm{K}$ *and only for one end of the rod.*

Karel combined formulas

Problem FoL.15 ... rubber bands

Consider two rubber bands (here, a rubber band is a single strip of rubber) with equal stiffnes
and equal rest length. One of the bands breaks if acted upon by a force greater than F_1. Th
other band breaks if acted upon by a force greater than cF_1, where $c > 1$ is a constant. W
ake a weight and hang it from both rubber bands in parallel so that they don't break. Then
ve slowly and continuously increase its weight (so that it does not oscillate) until the first ban
reaks. What's the minimal value of the constant c for which the second band doesn't brea
fterwards?

Michal was shooting rubber bands at people

Problem FoL.16 ... bungee jumping

There's a bridge of height $h = 50\,\mathrm{m}$ and a bungee jumping rope with rest length $l = 10\,\mathrm{m}$ an stiffness $k = 80\,\mathrm{N{\cdot}m^{-1}}$. A man with mass m attaches himself with this rope to the bridge an jumps off (consider his initial velocity to be zero). What's the largest possible mass of this ma such that the rope will stop him before hitting the surface below?

Michal is afraid to go bungee jumping

Problem FoL.17 ... fidget spinner

A fidget spinner rotates with angular frequency $\omega = 12.34\,\mathrm{s}^{-1}$. Inner and outer radii of the ba
bearing inside a fidget spinner are in ratio $k = 0.432$. What is the period of rotation of one ba
inside the bearing around the center of the toy? The inner part of the bearing is static. Assum
there is no slipping. *Matěj bought this autistic toy*

Problem FoL.18 ... insect in amber

Amber (with refractive index $n = 1.55$) is a transparent yellow-brown fossil resin. A sma beetle, trapped and preserved within the stone, appears to be 2.25 cm below the surface whe viewed directly from above. We are looking at the stone from a large distance. How far beneat the surface (in centimetres) is the beetle actually located?

Karel adapted a problem from Cutnell and Johnson: Physics 9e

Problem FoL.19 ... swung rod

There's a rigid rod with length $l = 1.2\,\mathrm{m}$ and zero mass, which is attached at one end in suc
a way that it can rotate around this fixed end. There are three small spheres with equal mas
ttached to this rod (you may consider them point masses). One of the spheres is at the fre
nd of the rod and the other two at $1/3$ and $2/3$ of its length. The rod is held horizontally a
irst. Then, we release it. What will be the velocity of its free end when it passes through th
quilibrium position (the bottommost point)? The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$

Karel was teaching mechanics

Problem FoL.20 ... deviated

There's a pendulum consisting of a thin, rigid rod and a heavy weight attached to the end o
he rod. The rod is then attached (by its other end) to another, horizontal rod. So, this secon
od represents the axis of rotation of the pendulum. Now we rotate the second rod so that i
s inclined at angle $\varphi = 30^\circ$ with respect to the horizontal plane (and it still represents th
xis of rotation of the pendulum). Find the period of small oscillations T' of the pendulum an
ompare it with the period T of a pendulum with a horizontal axis of rotation. Give the rati
T'/T as the result. *Mirek was watching the new series Genius – Einstein*

Problem FoL.21 ... slinger

We've got a slingshot made by attaching two ends of a massless rubber band to two points $l_0 =$
$= 15\,\mathrm{cm}$ *apart. Rest length of the rubber band is equal to this distance. We place a pebble wit*
mass $m = 5\,\mathrm{g}$ *in the middle of the rubber band and stretch it horizontally, forming the legs o*
n isosceles triangle with height $h_0 = 8\,\mathrm{cm}$*. Then, we release the pebble. What's the maximun*
elocity the pebble will reach? The rubber band (as a whole) has stiffness $k = 50\,\mathrm{kg{\cdot}s^{-2}}$*.*

Mirek was remembering his childhood toy

Problem FoL.22 ... energies of rotation

There are two identical homogeneous cylinders rotating with the same angular velocity ω. On ylinder, let's denote it A, is rotating around its main axis. The other cylinder B is rotatin round a parallel axis with distance $4R/5$ from the center of the cylinder, where R is the radiu f the cylinder. What's the ratio of rotational kinetic energies of the cylinders? We're intereste n E_B/E_A, where E_A and E_B are rotational kinetic energies of cylinders A and B, respectivel

Karel and Lukáš were riding on carouse

Problem FoL.23 ... the Ripper

Consider a barbell hanging horizontally in the air. For the purposes of this problem, we ca
imagine the barbell as a long thin rod with a small disk on each end. Let's charge one disk wit
charge $Q = 100\,\mu\mathrm{C}$*, the other disk with charge* $-Q$*; the rod is perfectly insulating and no charg*
can be transfered to the rod. Then, let's activate a homogeneous electric field with intensit
$E = 1\,\mathrm{MV{\cdot}m^{-1}}$ *parallel to the rod in the direction from negative to positive charge. Determin*
the stress (in units Pa*, with positive sign) in the middle of the rod caused by electrostati*
forces. The diameter of the rod is $d = 2\,\mathrm{cm}$*, the length of the rod* $l = 1\,\mathrm{m}$*. Neglect polarisatio*
of dielectrics.

Can't get ripped by lifting weights? Rip the weights

Problem FoL.24 ... inhomogeneous

Consider a method of launching rockets by imparting all the necessary momentum at th moment of launch. A rocket with mass $m = 10\,\mathrm{t}$ is launched directly upwards at its escap velocity. How high above the Earth's surface will the rocket be when its velocity drops to ha of the initial velocity? Express this result as a multiple of Earth's radius R (i.e. in units R) Neglect the effects of Earth's atmosphere and rotation.

Kuba meditating on the physics of balistic missiles

Problem FoL.25 ... Jáchym-style

Consider an isotropic point source of β radiation with activity $A = 3.4567\,\mathrm{MBq}$, located i a pool filled with a substance which dampens the radiation according to Lambert-Beer law $N(r) = N_0 \exp(-\mu r)$, where the absorption coefficient is $\mu = 1.1198 \cdot 10^{-1}\,\mathrm{m}^{-1}$. To wha distance r from the source should we place a Geiger-Müller counter with detection window are $S = 2.7183 \cdot 10^{-6}\,\mathrm{m}^2$, if we want it to detect $N = 10$ particles per second on average? Assum that the detector registers all particles which hit the detection window.

Lukáš didn't want to use a computer just to download problems

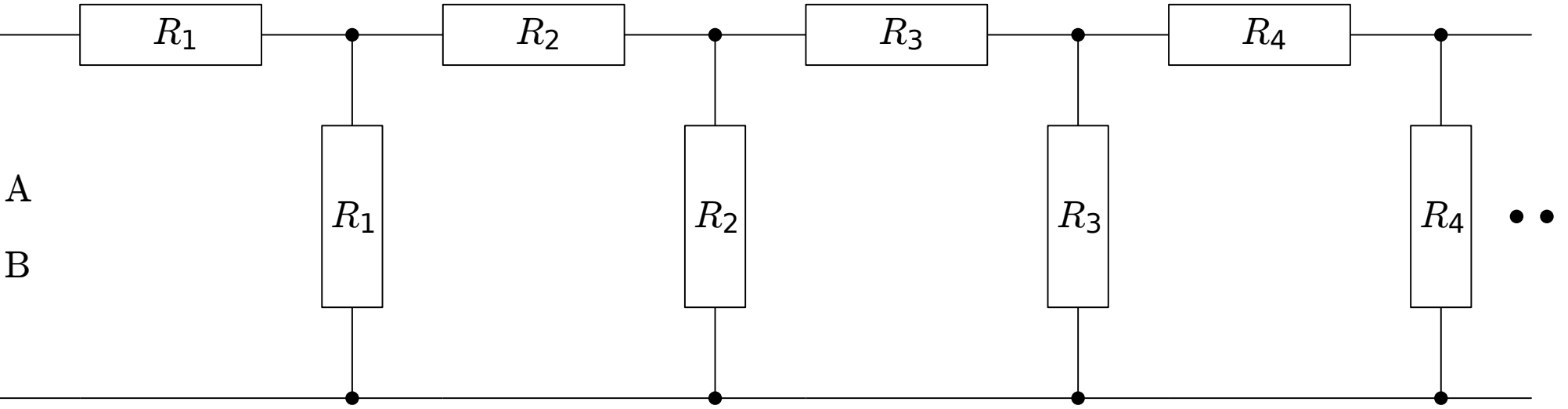
Problem FoL.26 ... drop the gallium into hot water

We've got $m_{\mathrm{Ga}} = 24\,\mathrm{g}$ *of gallium and we'd like to perform an interesting experiment with it We decide to prepare hot water with temperature* $t_0 = 93\,^{\circ}\mathrm{C}$ *and volume* $V = 250\,\mathrm{ml}$. *We plac he gallium into an imperfect calorimeter with heat capacity* $C = 95\,\mathrm{J{\cdot}K^{-1}}$ *and then pour th ot water on it. What will the temperature (in degrees Celsius) of the calorimeter, gallium an vater be after reaching thermodynamic equilibrium? Consider the calorimeter with water an*

gallium an isolated, closed system. The initial temperatures of the gallium and the calorimete were $t_1 = 22\,^\circ\mathrm{C}$. *The latent heat of fusion of gallium is* $l = 5.59\,\mathrm{kJ\cdot kg^{-1}}$, *specific heat capacit as a solid* $c_1 = 370\,\mathrm{J\cdot kg^{-1}\cdot K^{-1}}$ *and as a liquid* $c_2 = 400\,\mathrm{J\cdot kg^{-1}\cdot K^{-1}}$. *The specific heat capacit of water is* $c = 4{,}180\,\mathrm{J\cdot kg^{-1}\cdot K^{-1}}$.

Karel was thinking about the price of gallium, so he at least set this problem

Problem FoL.27 ... another infinite circuit?



What's the resistance between points A and B of an infinite resistor network in the figure
The resistors' resistances are $R_i = 2^{i-1}R$ – each pair of resistors has the same resistance, whic
s twice as large as that of the previous pair. Compute the numerical result for $R = 3.002\,\Omega$.

Karel was thinking about variations of a standard problem

Problem FoL.28 ... medal

The organizers of FYKOS decided to award the top three participants with medals. The FYKO medal is a flat cylinder consisting of three layers. The first layer is made of gold, the second laye is made of silver and the third one is copper. We know that the electrical resistance between th lower and upper base of the medal is the same as if all three layers were made of copper. We als know that the heat capacity of the medal is the same as if all three layers were made of gold Find out the mass ratio of the FYKOS medal and another medal of identicial proportions tha is made of silver only. Heat capacities, densities and resistivities of pure metals can be found o the Internet (or in printed engineering tables). Jáchym thinks that a diploma is not enough

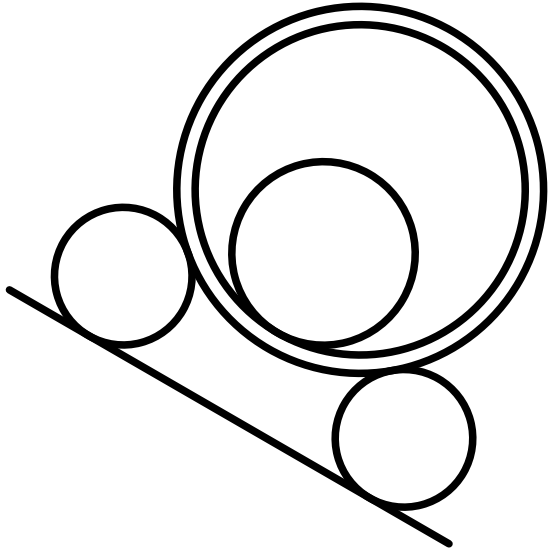
Problem FoL.29 ... Dyson sphere under construction

A Dyson sphere is a hypothetical construction surrounding a star, built by an advanced civili ation in order to utilise all energy coming from their star. It should be a relatively thin she with radius comparable to the distance of planets from the star. What would be the equilibriun emperature T_S of this shell compared to the equilibrium temperature T_P of a planet orbitin he same star on a circular orbit with distance equal to the radius of the Dyson sphere (i system without the Dyson sphere)? Assume that the planet and the Dyson sphere are blac odies and all bodies in this problem radiate isotropically. Neglect the cosmic background ra liation and the radiation of other space objects. As the result, compute the ratio $k = T_S/T_P$.

Karel was thinking about radiative heat transfe

Problem FoL.30 ... too many cylinders

On an inclined plane with inclination angle 35°*, there's a system of three full cylinders and one hollow cylinder. Two smaller cylinders have radius* $r_1 = 0.1\,\mathrm{m}$ *and moment of inertia* $J_1 = 2\,\mathrm{kg{\cdot}m^2}$*, the middle cylinder has* $r_2 = 0.15\,\mathrm{m}$ *and* $J_2 = 10\,\mathrm{kg{\cdot}m^2}$*, and the hollow cylinder has* $r_3 = 0.3\,\mathrm{m}$*,* $d = 0.02\,\mathrm{m}$ *and* $J_3 = 20\,\mathrm{kg{\cdot}m^2}$*, where* r_3 *is the outer radius and* d *is the thickness of its walls. All cylinders are homogeneous. A rigid construction of negligible mass keeps the cylinders in the same relative positions, but allows them to rotate. Assume there is no slipping anywhere. What will be the distance travelled by this system during the initial* $15\,\mathrm{s}$ *after being released from rest?*



Because one cylinder is too mainstream.

Problem FoL.31 ... do I hear that?

They say Chuck Norris can go get his beer so fast he meets himself. However, how would tha
work if Chuck couldn't break the laws of physics? According to the theory of relativity, he can'
move faster than light, so he can't see the photons he sent out (in vacuum, without reflectio
or refraction). However, could he hear himself? What's the minimum possible velocity wit
which he has to run if he wants to hear (faster, in reverse) what he said on the way there whe
running back? Chuck speaks with frequency $f = 200\,\mathrm{Hz}$ *and the human ear (Chuck's ear too*
can hear frequencies in the range $20\,\mathrm{Hz}$ *through* $20\,\mathrm{kHz}$*. The speed of sound is* $c = 340\,\mathrm{m{\cdot}s^{-1}}$*.*

Matěj imagined what it'd be like if Chuck Norris was subject to the laws of physics

Problem FoL.32 ... we'll be there in no time

In a galaxy far far away, a gigantic spaceship that can move with velocity $v = 0.002c$ was built. After reaching this velocity, the spaceship set course for Earth. What will be the error made by an observer on Earth when estimating its distance, if he believes the spaceship to be a star and measures its redshift as $z = 0.005$? Assume the Hubble constant is $H = 70\,\mathrm{km{\cdot}s^{-1}{\cdot}MPc^{-1}}$. Enter the result in **megaparsecs**.

Mirek was afraid of space invasion

Problem FoL.33 ... boring laboratory routine

We are sitting in a lab, measuring the spectral lines of hydrogen. One value of wavelength in th eries of measurements is extremely low, $\lambda = 91.184\,\mathrm{nm}$. *Assuming the Bohr model of hydroge s exact, what would be the distance of the electron whose transition caused this exceptiona mission? We also assume that the electron was in a bound state, no matter how large the initia listance was. In your calculations, use the following constants: ionization energy of hydroge* $E_0 = 13.598{,}4\,\mathrm{eV}$, *Planck constant* $h = 6.626{,}07 \cdot 10^{-34}\,\mathrm{J{\cdot}s}$, *speed of light* $c = 2.997{,}92 \cdot 10^{8}\,\mathrm{m{\cdot}s^{-}}$ *nd elementary charge* $e = 1.602{,}18 \cdot 10^{-19}\,\mathrm{C}$.

Hint *Use all the qunatities to the full given precision!* *Mirek sees atoms*

Problem FoL.34 ... loud music

At a disco, there's loud music blaring from one loudspeaker with power 200 W. *Matěj doesn'*
like that music though, so he decides to play his own on his phone via its speaker with powe
W. *How many people will hear Matěj's track louder than the disco? Matěj is standing* 10 m
from the loudspeaker and the number density of people is $2\,\mathrm{m}^{-2}$.

It's common to think about such things at a disco, right

Problem FoL.35 ... football player

What's the lowest velocity which a footballer has to impart to a ball located at the border o he penalty area in order to hit the crossbar? The distance to the goal is $d = 16.5\,\mathrm{m}$ *and th rossbar is at height* $h = 2.44\,\mathrm{m}$*. Neglect air resistance and the dimensions of the ball and th rossbar.*

Kuba wanted to cheat

Problem FoL.36 ... water over gold

Lord Waterboard wanted to have his waterproof castle filled with water. The drawbridge of th castle is rectangular (height $h = 3\,\mathrm{m}$ and width $s = 2\,\mathrm{m}$) and rotates around its bottom side The drawbridge is drawn up and locked at the top; the lock will hold up against forces up t $F = 50\,\mathrm{kN}$ (the hinges at the bottom can withstand any force). What will the water level i the castle be at the moment when the lock breaks and the water flows out and onto the poo villagers?

Matěj was thirsty

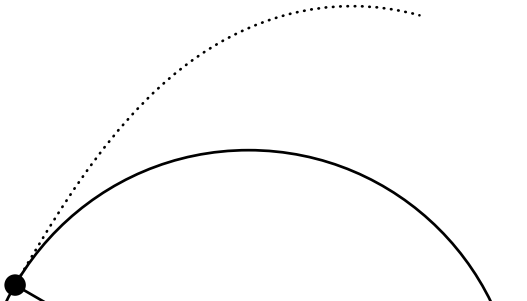
Problem FoL.37 ... unbent, unbowed, unsunken

On the flat water surface of a pond floats a hollow half-sphere with diameter $d = 30\,\mathrm{cm}$ *and mas* $m = 0.2\,\mathrm{kg}$. *By a vertical impulse the sphere is forced into oscillation. What is the maximun kinetic energy of the impulse that won't submerge the half-sphere? Neglect the resistive force of water.*

Xellos placed a spoon on his tea

Problem FoL.38 ... in the rain

You're riding a bike, it's raining and there's a puddle in front of you. You don't have any mudguards, so you slow down to $3\,\mathrm{m\cdot s^{-1}}$*. What will be the maximum height (in* cm *above ground) to which the water will splash when flying off a tyre with radius* 35 cm*? Neglect all resistive forces.* *Štěpán rode a bike in the rain.*



Problem FoL.39 ... fire up the synapses

The magnetic field of Earth suddenly vanished and we need to replace it somehow. We know that the original field was approximately like that of a dipole and its magnitude at the equator is approximately given by the formula $|B| = B_0/R^3$, where $B_0 = 3.1 \cdot 10^{-5}\,\mathrm{T}$ and R is the distance from the center of Earth as a multiple of Earth's radius. We want to replace the field using a small coil (solenoid) inserted into the center of Earth. The coil has $N = 10^6$ loops and its radius is $\varrho = 1\,\mathrm{m}$. What current should flow through the coil if the new magnetic field is to be equal to the initial field? Assume that the permeability of all materials is the same as that of vacuum.

Mirek cannot count to four

Problem FoL.40 ... efficient ride

An electric car is driving on a level road. Its effective cross-section is $S = 2\,\mathrm{m}^2$ and its dra coefficient is $C = 0.2$. Its motor has efficiency $\eta = 40\,\%$. What's the optimal constant velocit (which maximises the car's range), if the power consumed inside the car (by air conditioning radio,...) is a constant $P_0 = 400\,\mathrm{W}$? The density of air is $\varrho = 1.29\,\mathrm{kg{\cdot}m^{-3}}$.

Matěj dreams of driving a Tesl

Problem FoL.41 ... hot day

During the spring equinox a little boat is swimming in the equatorial seas. The boat ha
he shape of a rectangle with surface area $5\,\mathrm{m}^2$*. The Sun shines on the surface of Earth wit*
ntensity $1.3\,\mathrm{kW{\cdot}m^{-2}}$*. How much solar energy in* MJ *will the boat receive from sunrise t*
unset?

Štěpán gazed on hot metal roofs

Problem FoL.42 ... all the way up, please

What's the minimum length (in kilometres) of a space elevator built at the equator, so that i
doesn't collapse under gravity? Consider the elevator to be just a few straight ropes leading t
pace. The elevator is homogeneous and doesn't contain any extra weight at the end.

We're not interested in the trivial solution of zero length. *Štěpán forgot how to use a lif*

Problem FoL.43 ... shower

What's the maximum useful running time for a shower fed from a cylindrical boiler with cross ection $S = 0.8\,\mathrm{m}^2$ and height $H = 1.5\,\mathrm{m}$? The showerhead is connected directly to the botton of the boiler, the area through which water flows out of the showerhead is $s = 0.8\,\mathrm{cm}^2$. Conside he minimum volumetric flow rate for which showering is possible to be $Q_0 = 2\,\mathrm{dl{\cdot}s}^{-1}$. Assum hat there's no water flowing into the boiler and all water flowing into the showerhead come rom the boiler.

Enter the result in minutes. *There wasn't enough water for Kuba*

Problem FoL.44 ... space race

Two spacehips, USS Annie and USS Bonnie, are racing. Both are approaching the finishin
ine with their maximum speeds $v_A = c/4$ and $v_B = c/2$. An observer at rest at the finishin
ine at time 0 observes Bonnie at distance $d_B = 250\,\mathrm{km}$ and Annie at distance $d_A = 100\,\mathrm{km}$
Determine the absolute difference (positive number) between the times when the spaceship
each the finishing line according to this observer in ***milliseconds****.*

According to Mirek, placings in a race are relative

Problem FoL.45 ... gray plates

There are two parallel infinite planes in vacuum. The first plane has fixed temperature T_1 =
$= 200\,\mathrm{K}$ *and reflectivity* $R_1 = 1/2$ *(the ratio of reflected to incident light intensity). The secon*
plane has fixed temperature $T_2 = 300\,\mathrm{K}$ *and reflectivity* $R_2 = 1/3$*. What will be the temperatur*
of a third plane with reflectivity $R = 1/3$ *placed in parallel between them?*

Assume that all planes radiate as black bodies (have emissivity 1), they just partially reflec
ncident light. The transmissivities of all bodies are zero.

Kuba wanted to quantify the cooling effect of reflectio

Problem FoL.46 ... do not drown!

There's a river of width $d = 20\,\mathrm{m}$ with a parabolic velocity profile, meaning that at the distanc
from either bank, the speed of the current is $v(y) = 4v_0 \frac{y}{d}\left(1 - \frac{y}{d}\right)$, where $v_0 = 1\,\mathrm{m{\cdot}s^{-1}}$ is th
peed in the middle. The vertical velocity profile is constant. How long would it take us to swi
cross the river if we stay on a trajectory perpendicular to the bank? In the absence of an
urrent our swimming speed would be $w = 2v_0$. *Kuba was carried away by the current*

Problem FoL.47 ... pressure cooking

A pressure cooker is a closed pot with walls of thickness $t = 5\,\mathrm{mm}$*, thermal conductivity* $\lambda =$ $= 9\,\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$*, volume* $V = 3\,\mathrm{l}$ *and surface area* $S = 4\,\mathrm{dm}^2$*. There is a single hole in its wall with diameter* $d = 4\,\mathrm{mm}$*. We start heating the pot with power* $P = 7\,\mathrm{kW}$*, pour* $V_\mathrm{v} = 1\,\mathrm{l}$ *c water inside, close the lid and bring it boil. The room temperature is* $T_\mathrm{i} = 20\,^\circ\mathrm{C}$*. Neglect th dependence of boiling point on pressure. What will be the pressure increase (with respect t atmospheric pressure) inside the pot?*

Xellos knows how to cook water

Problem FoL.48 ... thirteen barrels

Consider a hermetically sealed container with known volume containing a gas with known pressure. We also have a barrel with volume equal to one thirteenth of the volume of the container. First, we connect the barrel to a source of gas (which fills it with gas with fixed pressure) and wait for the pressure in the barrel to equalise. Then, we disconnect the barrel from the source, connect it to the container and wait for the pressures to equalise. Finally, we disconnect the barrel from the container. What must the ratio of pressure of gas in the source to initial pressure in the container be if we want to repeat this process exactly 13 times in order to increase the pressure in the container to exactly 13 times its initial value? Assume that the ambient temperature is constant and that both the barrel and the container are in thermal contact with their surroundings.

Jáchyma got an idea on Friday the 13th

Problem FoL.49 ... space cowboy

An astronaut with a space gun is floating freely in empty space. The gun is a futuristic weapon hat, when fired, imparts the projectile with kinetic energy $E_\mathrm{k} = 1 \cdot 10^{17}\,\mathrm{J}$ *(measured in th eference frame in which the center of mass of the gun and projectile is at rest). Find th elocity with which the projectile will move away from the astronaut (in the reference fram f the astronaut). The mass of the astronaut with the gun is* $m_1 = 100\,\mathrm{kg}$*, the mass of th rojectile is* $m_2 = 10\,\mathrm{kg}$*. Neglect any decrease in mass due to burning of explosive charge nd rotation of objects.* ***Enter the result as a multiple of the speed of light (in units o).***

Mirek thinks that guns are operational in vacuum

Problem FoL.50 ... pressure cooking reloaded

A pressure cooker is a closed pot with walls of thickness $t = 5\,\mathrm{mm}$*, thermal conductivity* $\lambda =$ $= 9\,\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$*, volume* $V = 3\,\mathrm{l}$ *and surface area* $S = 4\,\mathrm{dm^2}$*. There is a single hole in its wall with diameter* $d = 4\,\mathrm{mm}$*. We start heating the pot with power* $P = 7\,\mathrm{kW}$*, pour* $V_\mathrm{v} = 1\,\mathrm{l}$ *of water inside close the lid and bring it to boil. The room temperature is* $T_\mathrm{i} = 20\,^\circ\mathrm{C}$*. Compute the increase in the boiling point of water (in* $^\circ\mathrm{C}$*) with respect to the boiling point at standard conditions (in an open pot), which is* $100\,^\circ\mathrm{C}$*.*

Xellos knows how to cook water

Problem FoL.51 ... balls of steel

There are two conducting balls with radii $R_1 = 0.1\,\mathrm{m}$, $R_2 = 0.2\,\mathrm{m}$ *inside a homogeneous electri*
ield $\boldsymbol{E} = E\boldsymbol{e}_z$, *where* $E = 50\,\mathrm{kV{\cdot}m^{-1}}$. *The spherical coordinates of the center of the second ba*
with respect to the center of the first one are (r, ϑ, φ), *where the vector corresponding to* $\vartheta =$
ies in the direction of cartesian axis z *and* $0 \leq \varphi < 360°$. *Find the force between those ball*
or $R_{1,2} \ll r = 5\,\mathrm{m}$, $\vartheta = 30°$, $\varphi = 50°$. *Xellos wanted to make a problem that has balls*

Problem M.1 ... tic-toc

How many times greater is the distance travelled by the tip of the second hand on a clock fac
than the distance travelled by the tip of the minute hand, during one day (exactly 24 hours)
The length of the second hand is 105 mm, *of the minute hand* 100 mm.

Kiki was killing some time

Problem M.2 ... far from school

Luboš and Náry leave the student dormitory at the same time and both head (separately) t
chool. The distance between the school and the dormitory is $l = 350\,\mathrm{m}$*. Luboš has got a swif*
pace, $v_\mathrm{L} = 2\,\mathrm{m{\cdot}s^{-1}}$*, but he is also very forgetful. During his walk to school he had to retur*
hree times to the dormitory – the turning points were at distances $100\,\mathrm{m}$*,* $200\,\mathrm{m}$ *and* $300\,\mathrm{m}$
rom the dormitory. Each time he spent $t_\mathrm{L} = 1\,\mathrm{min}$ *inside the dormitory. Náry walks at th*
ame speed as Luboš, $v_\mathrm{N} = v_\mathrm{L}$*, but he doesn't turn back to get the things he forgot. Howeve*
e has another bad habit – stopping and talking to passersby. Let's assume he bumps int
omeone every $50\,\mathrm{m}$ *(including the dormitory entrance and school entrance) and talks to hi*
or $t_\mathrm{N} = 1.5\,\mathrm{min}$*. What will be the difference of Luboš's and Náry's travel times? If Náry arrive*
irst, give the result as a negative number. *Mirek counted how many times Luboš returned*

Problem M.3 ... not falling well

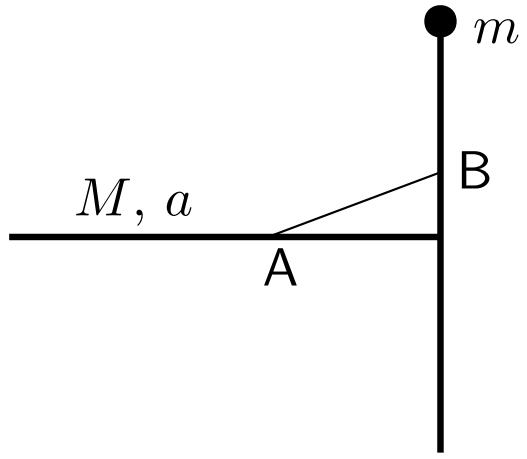
A cylindrical glass with radius of the base $R = 5\,\mathrm{cm}$ is filled with beer so that the surface o the liquid reaches $h = 4\,\mathrm{mm}$ above the edge (measured at the center of the surface). This is du to the strong surface tension of $\sigma = 72\,\mathrm{mN{\cdot}m^{-1}}$. A fly just drowned in the beer and floats a the distance $r = 2\,\mathrm{mm}$ from the center of the surface. How fast will the fly be moving when i reaches the edge of the glass? Assume it was initially at rest, the slope of the surface close t the center is small (but non-zero) and all resistive forces are negligible.

Xellos was drinking a beer

Problem M.4 ... don't fall at all

n the figure there is a construction made of two rods as seen from bove. The length and mass of each rod are a and M. A mass $m = = M/4$ is attached to the end of one of the rods. We have welded nother homogeneous rod to the construction (the thin line in the igure) so that the center of mass of the new rod coincides with he center of mass of the construction. Find the length of the third od (distance between A *and* B *in the figure) and give the result as multiple of a.*

Mirek was unhappy because he couldn't touch the center of mass.



Problem E.1 ... breaking a capacitor

We've got a capacitor formed by two separated conducting plates. We charge it and disconnec
t from the voltage source. How many times larger will the charge we can get from the capacito
pe, if we break it into four identical pieces and place them on top of each other without rotatin
hem in any way? *Štěpán was breaking chocolate into pieces*

Problem E.2 ... ideal voltage

Consider an ideal voltage source. First, we connect to it in parallel three identical resistor and find out that the total power consumed by the resistors is P. What will be the consume power if we connect the resistors to the source in series? Compute the result as a multiple o P.

Karel was teaching about electric voltage

Problem E.3 ... ideal current

Consider an ideal current source. First, we connect to it three identical resistors in parallel and find that the total power consumed by the resistors is P. What will the consumed power be if we connect the resistors to the source in series? Compute the result as a multiple of P.

Karel was teaching about electric current

Problem E.4 ... loaded triangle

We built a triangle from three resistors with equal resistances and connected voltage source
parallel to two of the resistors in such a way that their negative poles are connected to the sam
vertex. How many times larger will the current flowing through the source with voltage $7U$ b
compared to the current flowing through the other source with voltage $5U$?

Kuba forced Štěpán to make some problems instead of programming

Problem X.1 ... a nice cup of tea

Somebody mixed the polonium-210 isotope (half-life 138 days*) into Mikuláš's tea. Luckily Mikuláš noticed that. However, he really wants some tea, so he measured the amount of ra dionuclide in his tea and calculated when will the radioactivity decrease to a safe-to-drink value After this time of* 342 days *he drank his tea with no side effects. A few years later, somebod mixed twice the original amount of polonium-210 into his tea. How long does Mikuláš have t wait this time?*

Guys, I really am not a retired Russian agent. .

Problem X.2 ... cheesy

Vítek found a shop with suspiciously cheap cheese. On Monday at 14:15, he bought 250 g *cheese there. On Tuesday at 17:21, he bought another* 200 g *and on Wednesday at 19:45, yet another* 160 g*. He always stored the cheese he bought carefully in the fridge. Unfortunately, he hadn't noticed that the cheese contained the radioactive nuclide* ^{24}Na *with half-life* 15 hrs

The mass fraction of this substance in the cheese at the moment when the cheese is bought i
$v = 0.0013$. *What will be the mass (in grams) of non-decayed* ^{24}Na *in the cheese at the momen*
when Vítek wants to eat it – on Thursday at 9:11? Jáchym was skipping through sale flyers

Problem X.3 ... yoghurt

Jáchym eats half a kilogram of yoghurt for dinner every day. The yoghurt contains a mas fraction $w = 10^{-4}$ of a certain radioactive isotope with half-life 26 days. How many grams o this isotope will be present in Jáchym's body every day before dinner, if he's been eating thi way for a long time? Assume that the isotope doesn't leave his body in any way other tha through radioactive decay.

Jáchym, eating a yoghurt

Problem X.4 ... beer

The average volume ratio of air to beer in beer foam is 6.87. *The half-life of an average bubbl*
s 73 s. *If, initially, the beer reaches the height* 13 cm *above the bottom of the glass and the foa*
eaches height 16.5 cm, *what height (in centimetres) will the foam reach after two minutes?*

Made up by Jáchym while watching a Czech cult film